

RELEASE & IMPLEMENTATION SUMMARY

SKYNN AI v2.1 — beta

One product, two experiences: the Basic AI Skin Analysis and the Advanced AI Dermatology Analysis.

DATE 28 September 2026	CODE PR #161 · branch <code>claude/skynn-ai-v2-1-audit-s45htw</code>	DATABASE Supabase project <code>gnkpzijxuciiaamakgzm</code>
STATUS Release candidate — pending merge	AUDIENCE Internal team, partners, diligence	CONTAINS No credentials, no personal user data

1. Executive summary

SKYNN AI v2.1 — beta consolidates SkinLabs' skin-analysis surfaces into one coherent product with two clearly separated experiences:

- **Basic AI Skin Analysis.** A free, deterministic analysis of the member's answers. Glow Explorer and Glow Lite get one per rolling 7 days; Glow Insider and VIP get unlimited analyses.
- **Advanced AI Dermatology Analysis.** A longer questionnaire that requires an Analysis Pass. During this beta it runs as a secure pending-submission (fallback) workflow while the clinically approved report pipeline is finalised.

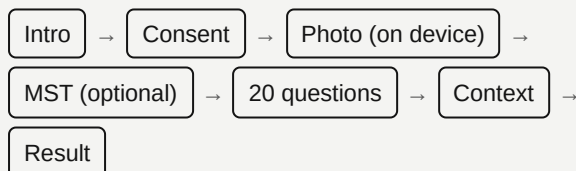
The release was run as an audit first. It found:

- a legacy path that sent members' photos to a model and asked it to estimate skin tone;
- a second, ungated "Advanced" flow;
- an exploitable Analysis Pass refund bypass;
- a Basic PDF that claimed dermatologist review that never happened;
- six competing names for the same product.

All of these are fixed. The database side is applied and verified on production. The frontend ships when PR #161 is merged.

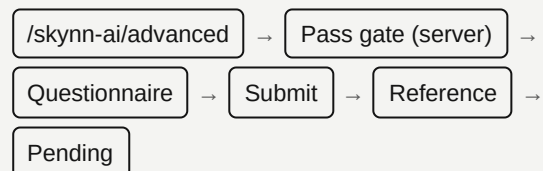
2. Product architecture

Basic AI Skin Analysis



- A rule-based engine in the browser: no model call and no photo processing.
- The full result and PDF appear only after `save_starter_analysis()` accepts the save. That call is also where the 7-day limit is enforced.
- Anonymous visitors see a preview (skin type and top concerns) and sign up to save.
- A Basic AI Skin Analysis never spends an Analysis Pass.

Advanced AI Dermatology Analysis



- Access comes only from `get_advanced_assessment_access()`. Membership alone never qualifies.
- One Pass is consumed inside `submit_advanced_assessment_session()`. A retry returns the same submission, and a second open submission is refused before any charge.
- Every Advanced CTA on the site now leads to this single flow.

3. Terminology standardisation

CANONICAL

REPLACES (CURRENT PRODUCT COPY)

SKYNN AI v2.1 — beta	SKYNN AI (beta), Skynn AI, SKYNN.AI
Basic AI Skin Analysis	Starter Analysis, Starter Skin Analysis, free starter analysis
Advanced AI Dermatology Analysis	Advanced Assessment, Advanced Analysis, Advanced Skin Analysis, Advanced (AI) Dermatology Report, SKYNN AI Advanced, Advanced AI Analysis
... submission / — Pending	Wording for the fallback record. It is never called a report, reviewed, approved or released.
Analysis Pass	Inconsistent capitalisation
Monk Skin Tone (MST)	Always described as optional, self-reported, never inferred from a photo, and not a diagnosis
SKYNN AI	AI Formulator. <code>/ai-formulator</code> remains as a redirect only.

Intentionally kept:

- Database identifiers: `save_starter_analysis`, `starter_analyses_used`, `advanced_assessment_*`.
- Historical analytics event names.
- The FAQ slug.
- Dated historical content: announcements, the About roadmap and newsroom articles.

The mapping is in `docs/skynn-terminology.md`.

Validation:

- A static test scans the UI, data, SEO config, email templates, intake PDF and API errors, and fails on any retired name. It also checks the frontend and edge-function copies of the terminology constants stay identical.
- An end-to-end test reads the rendered text of `/skynn-ai`, `/skynn-ai/advanced` and `/routines` on desktop and mobile, in light and dark themes.

4. Major changes

SEV.	FINDING	FIX
P0	The legacy <code>skincare-ai</code> path sent members' photos to a model with a "Fitzpatrick estimate" instruction.	The live function was patched the same day (no photo, no tone instructions). Its source is now a retired 410 stub, which is deployed after merge.
P0	A second "Advanced" flow ran inside <code>/skynn-ai</code> . It let members get a result without a Pass, charged Explorers a Pass for a free-tier model, and skipped the review hold and reference number.	Removed. Every Advanced CTA goes to <code>/skynn-ai/advanced</code> , and membership no longer implies access (entitlements, <code>/routines</code> , FAQ and plan copy).
P0	A member could refund the Pass spent on their own Advanced submission with <code>refund_analysis_pass()</code> .	Reproduced on production in a rolled-back probe, then fixed and re-verified. 0 real sessions were affected.
P0	The Basic PDF stated "dermatologist-reviewed" and "reviewed by SkinLabs specialists". Neither was true.	It now states that the analysis is rule-based, not reviewed by a dermatologist, that the photo is not analysed, and that it is not medical advice.
P0	Anonymous visitors got the full result and an automatic PDF, with no server-side limit.	They now see a preview only. The full result and PDF appear only after a successful server save.
P1	A Basic save fell back to spending an Analysis Pass, and an account without a profile row got unlimited free saves.	Both removed and tested.

P1	Analysis Pass consumption had no lock.	A per-user transaction advisory lock was added to consume and refund.
P1	Photos were uploaded and never read, alongside an unimplemented "deleted within 30 days" promise.	Photos are no longer uploaded, and the copy is now accurate. Production held 0 stored Basic photos. Bucket size and MIME limits were added.
P1	MST copy said MST didn't affect recommendations, which is false. The database also allowed <code>model_estimated</code> .	Copy corrected. The CHECK constraint now allows <code>user_reported</code> only.
P1	A Pass holder was never shown the weekly lock and only hit it after the quiz. This was found by the e2e tests.	Fixed in <code>useFormulatorAllowance</code> .
P1	On phones, the new PDF button overflowed its card. This was found in the browser walkthrough.	Fixed, and an overflow assertion was added to the e2e suite.
P2	Two competing SEO title sources for <code>/skynn-ai</code> .	There is now a single source (<code>seo-config</code>), with JSON-LD naming the Basic AI Skin Analysis.

5. Advanced AI Dermatology Analysis — fallback workflow



- No model is called and no Anthropic key is required. Fallback rows cannot be claimed by the production worker, as the probe verifies.
- The intake PDF is headed "Advanced AI Dermatology Analysis — submission — intake record (not a report) · STATUS: PENDING". It carries a Release row "SKYNN AI v2.1 — beta" and the stored version snapshot: prompt set, questionnaire, scoring and evidence versions.
- A failure in the email or PDF step never loses the submission. The worker retries with backoff, and admins can retry, reject with a refund, or promote once production mode exists.
- The future production report requires a recorded dermatologist sign-off and a working AI key, and every report is held for admin review. **No clinical approval is claimed in this release.**

6. Safety and privacy

- **No photo-based skin-tone inference.** Photos stay on the device, and no SKYNN AI edge function sends an image to a model. A test scans the SKYNN edge code for image-to-model calls and tone-estimation prompts.
- **MST** is optional and self-reported, stored only as `user_reported`, never guessed, and not a diagnosis.
- **POPIA:** there is explicit consent for the Basic analysis and explicit special-information and cross-border consent for the Advanced analysis, recorded before any Pass is consumed. Intake PDFs sit in a private bucket with no client policies. Members can withdraw and delete a submission, which refunds the Pass and alerts `reports@` to delete the mailbox copy.
- **Server-side trust only:** entitlement, Pass balance, submission status and the weekly limit are decided in SQL, never by the browser. Row-level security isolates each member's sessions and submissions, as the probe verifies.

7. Analytics

All funnel events go through `trackSkynnEvent()`, which applies a whitelist before anything is sent. Only mode, step, progress milestone, booleans, a safe error category, source and feature version pass through. Answers, MST values, free text, images and reference numbers never leave the browser. Events distinguish `mode: basic` from `mode: advanced`.

STAGE	EVENTS
Discovery	<code>skynn_viewed</code> · <code>skynn_started</code> · <code>skynn_mode_selected</code>
Consent / photo / MST	<code>skynn_consent_*</code> · <code>skynn_photo_*</code> · <code>skynn_mst_*</code>
Questionnaire	<code>skynn_assessment_started</code> · <code>skynn_questionnaire_progress</code> (25/50/75) · <code>skynn_questionnaire_completed</code>
Basic	<code>skynn_basic_submission_*</code> · <code>skynn_basic_limit_reached</code>
Advanced	<code>skynn_advanced_entitlement_checked</code> · <code>skynn_analysis_pass_required/confirmed</code> · <code>skynn_advanced_submission_*</code> · <code>skynn_advanced_pending</code> · <code>skynn_advanced_reference_created</code>
Results / errors	<code>skynn_results_viewed/completed</code> · <code>skynn_results_pdf_*</code> · <code>skynn_error</code>

Admin → SKYNN Reviews has an operations panel (`skynn_ops_summary()`, admin-only, counts only). It shows:

- Basic analyses saved and weekly-limit hits;
- starts, results viewed, PDF downloads and errors;
- Advanced submissions and pending count;
- intake PDF and email failures;
- Passes used.

8. Database and backend

OBJECT	CHANGE	STATE
<code>20260928140000_skynn_v21_basic_limit_and_hardening</code>	7-day window; <code>save_starter_analysis</code> (no Pass fallback, <code>profile_missing</code>); <code>consume_analysis_pass</code> lock; bucket limits; MST CHECK; <code>skynn_ops_summary</code>	Applied live
<code>20260928141000_skynn_v21_refund_guard</code>	<code>refund_analysis_pass</code> refuses session-linked or >15-minute-old transactions; lock	Applied live
<code>20260928142000_skynn_v21_plan_copy</code>	<code>pricing_plans.benefits</code> copy (Basic every 7 days / unlimited)	Applied live
<code>20260928150000_skynn_v21_retire_legacy_ai</code>	Revokes legacy client grants; drops the client insert policy	After merge

Edge: <code>skincare-ai</code>	v36 interim (no photo, no tone inference) → 410 stub	Stub after merge
Edge: <code>email-processor v38</code> , <code>skynn-advanced-worker v24</code> , <code>skynn-advanced-assessment v34</code>	New naming in emails, intake PDF, internal email and API errors	Deployed, pinned to <code>e7e3389</code>

9. Quality assurance (verified)

CHECK	RESULT
TypeScript (<code>tsc -p tsconfig.app.json</code>)	0 errors
ESLint	0 errors. 22 warnings, all in files this release doesn't touch.
Unit tests (<code>bun test</code>)	432 tests, 0 failures. Includes the terminology guard, the no-tone-inference guard, the analytics whitelist and the 7-day limits.
End-to-end (Playwright, 4 projects: desktop and mobile × light and dark)	56 passed, 4 skipped. The skipped tests are the PayFast sandbox journeys, which need a sandbox merchant. The SKYNN journeys cover: - anonymous preview without a PDF; - signed-in save, then the full result and PDF, with no horizontal overflow; - the 7-day lock showing the next date; - Advanced routing and the Pass gate; - Pass → submit → pending reference; - no retired names; - the <code>/ai-formulator</code> redirect; - no calls to <code>skincare-ai</code> or the legacy RPCs.
SQL probes on production (rolled back)	Basic allowance: 19/19. Advanced Pass gate and row-level security: 22/22.
Browser walkthrough (production build, mocked backend)	Screenshots at phone and desktop sizes in light and dark: intro, consent, photo, MST, results with PDF, Advanced landing, review and pending confirmation. One mobile overflow was found and fixed.
Live Advanced end-to-end (production, synthetic submission on the admin account)	- Reference <code>SKYNN-ADV-20260928-YPH353</code> issued and the Pass consumed. - The worker generated a 30.6 KB intake PDF on its first attempt. - The internal email to <code>reports@</code> was handed off, and the member confirmation email was accepted by the email provider (a provider message ID was returned). - The test submission was then deleted through the member-delete path, and its Pass was refunded. Inbox delivery was not independently confirmed.
Security advisor	No new findings beyond the repository's existing "authenticated can execute SECURITY DEFINER" pattern. The new admin function is gated internally.

10. Known limitations

- The Basic engine runs in the browser, which is inherent to a deterministic engine. The server enforces the saved result and the weekly limit, not the computation.
- The retirement of the legacy grants and the `skincare-ai` 410 stub wait for the merge, because the current production frontend still uses them. Until then the deployed `skincare-ai` is the interim version, which never receives a photo or asks for skin tone.
- The Privacy Policy, Terms and Whitepaper still describe photo upload and analysis. This now over-describes what happens, and the legal wording is left for a human decision.
- `/routines` markets "Smart Routines", which isn't built yet. That is a pre-existing issue outside SKYNN AI.
- The Advanced production report pipeline (dermatologist sign-off record plus a working AI key) is not active. Fallback mode is what's live.
- The Vercel preview build of PR #161 was not inspected from this environment.

11. Release status

SKYNN AI v2.1 — beta — Release Candidate Verified (pending merge)

Every P0 and P1 item above is fixed and verified. Build checks, unit tests, e2e tests, SQL probes and the live Advanced run pass. It is not yet live for users: the frontend ships when PR #161 is merged. The two post-merge steps (the `skincare-ai` 410 stub and the legacy-grant revoke migration) must follow immediately after. No clinical approval is claimed.

SkinLabs® · SKYNN AI v2.1 — beta · Release & Implementation Summary · 28 September 2026. Generated from the repository and live verification results. The code and database are the source of truth if this document ever disagrees.